Muon and Spectral Optimizers

When matrix geometry changes the effective covariance

Princeton 2026 Summer School in Machine Learning

Covariance now acts on a matrix gradient

The first three lectures treated power-law covariance as a structure on directions. For a weight matrix, the minibatch gradient G is itself a matrix. If

$$G = U\,\mathrm{diag}(\sigma_1, \ldots, \sigma_r)V^\top,$$
$$\boxed{\mathrm{SignSVD}(G) = UV^\top.}$$

SignSVD keeps the singular vectors and replaces every nonzero singular value by one. Muon approximates this operation with a few matrix multiplications.

How do covariance and batch size control gradient orthogonalization?

Backpropagation produces an outer product

Let

$$\mathcal{L}(\theta) = \mathbb{E}_{(X,Y)}\left[\ell(F_\theta(X), Y)\right].$$

For one layer, write $z = Wa$ and $\delta = \nabla_z \ell(F_\theta(X), Y)$. A perturbation dW gives

$$d\ell = \sum_{i,j} \delta_i a_j \, dW_{ij} = \langle \delta \otimes a, dW \rangle_F,$$

$$\implies \quad \boxed{\nabla_W \ell = \delta \otimes a.}$$

A batch gradient has rank at most the batch size

For a minibatch of B examples,

$$\boxed{G_B = \frac{1}{B}\sum_{b=1}^{B} \delta_b \otimes a_b, \qquad \operatorname{rank}(G_B) \le B.}$$

Right geometry

The activation covariance acts on the input side of W.

Left geometry

The backpropagated-error covariance acts on the output side.

Batch size limits the number of matrix directions visible to a spectral update.

A bilinear regression model preserves the matrix geometry

Take independent

$$x \sim \mathcal{N}(0, \Sigma_{\mathrm{in}}), \qquad y \sim \mathcal{N}(0, \Sigma_{\mathrm{out}}),$$

and let $\Delta = W - W_\star$. The single-sample loss is

$$\ell(W; x, y) = \frac{1}{2}\langle y \otimes x, \Delta\rangle_F^2.$$

Its gradient and population Hessian are

$$\boxed{\begin{aligned} \nabla_W \ell &= (y \otimes x)\langle y \otimes x, \Delta\rangle_F, \\ \mathcal{K}(H) &= \Sigma_{\mathrm{out}} H \Sigma_{\mathrm{in}}. \end{aligned}}$$

This is ordinary least squares whose matrix layout remains visible.

Two covariances create N^2 paired modes

In the two covariance eigenbases,

$$\Sigma_{\mathrm{out}} u_i = \mu_i u_i, \qquad \Sigma_{\mathrm{in}} v_j = \lambda_j v_j,$$
$$\mathcal{K}(u_i \otimes v_j) = \mu_i \lambda_j (u_i \otimes v_j),$$
$$\Delta = \sum_{i,j} a_{ij} (u_i \otimes v_j).$$

Consequently,

$$\boxed{\mathcal{R}(W) = \frac{1}{2} \sum_{i,j} \mu_i \lambda_j a_{ij}^2.}$$

The eigenbasis is indexed by pairs (i, j), so an $N \times N$ matrix naturally has N^2 spectral order parameters.

A spectral optimizer reshapes singular values

update	matrix operation	action on σ
SGD	$U\Sigma V^\top$	$\sigma \mapsto \sigma$; preserves the gradient hierarchy
power p	$U\Sigma^p V^\top$	$\sigma \mapsto \sigma^p$; interpolates or reverses the hierarchy
SignSVD	$UV^\top$	$\sigma \mapsto 1$ on the nonzero subspace
instant Shampoo	$(GG^\top)^{\dagger/4}G(G^\top G)^{\dagger/4}$	also $\sigma \mapsto 1$
finite-NS Muon	$\mathrm{NS}_K(G)$	rapidly moves σ toward one

Full flattening is one point on a broader spectral-power design axis.

SignSVD is the polar factor

For $G = U\Sigma V^\top$,

$$\boxed{\mathrm{polar}(G) = (GG^\top)^{\dagger/2} G = UV^\top.}$$

It is basis-covariant:

$$\mathrm{polar}(QGR^\top) = Q\,\mathrm{polar}(G)\,R^\top.$$

Entrywise SignSGD instead uses

$$[\mathrm{sign}(G)]_{ij} = \mathrm{sign}(G_{ij}),$$

so it depends on the coordinate basis.

Muon separates a temporal choice from a spectral choice

A stylized Muon update is

$$M_{t+1} = \beta_{\mathrm{mom}} M_t + G_t,$$
$$W_{t+1} = W_t - \eta \, \mathsf{NS}_K(M_{t+1}).$$

It adds two operations to exact SignSVD:

1. average gradients through momentum;
2. approximate the polar factor without computing an SVD.

Instantaneous Shampoo gives the same polar factor because

$$(GG^\top)^{\dagger/4} G (G^\top G)^{\dagger/4} = UV^\top.$$

Historical context: Carlson–Cevher–Carin (2015); Tuddenham–Prügel-Bennett–Hare (2022); Gupta–Koren–Singer (2018).

Newton–Schulz maps singular values toward one

Start with the cheap Frobenius normalization

$$X_0 = \frac{G}{\|G\|_F + \varepsilon}, \qquad \|X_0\|_{\mathrm{op}} \le 1.$$

The classical polar iteration is

$$X_{k+1} = \frac{1}{2}(3X_k - X_k X_k^\top X_k), \qquad s_{k+1} = \frac{1}{2}(3s_k - s_k^3).$$

Muon commonly uses five steps of

$$X_{k+1} = aX_k + (bA_k + cA_k^2)X_k, \qquad A_k = X_k X_k^\top, \qquad (a, b, c) \approx (3.44, -4.78, 2.03).$$

The large slope accelerates sharply near zero.

The clean model isolates the spectral operation

From now on:

$$\boxed{\beta_{\mathrm{mom}} = 0, \qquad \mathsf{NS}_K(G) \longrightarrow \mathsf{polar}(G).}$$

The resulting algorithm is exact SignSVD.

This removes two separate approximation questions:

- no temporal averaging of successive gradients;
- no finite polynomial error in the polar map.

Momentum returns only after the covariance and batch-size effects are understood.

A matrix update still splits into descent and volatility

One step is descent plus volatility

For residuals $d_b = \langle y_b \otimes x_b, \Delta \rangle_F$, define

$$G(\Delta) = \frac{1}{B} \sum_{b=1}^{B} d_b (y_b \otimes x_b), \qquad S = \mathrm{polar}(G),$$
$$a_{ij} = u_i^\top \Delta v_j, \qquad s_{ij} = u_i^\top S v_j, \qquad q_{ij} = a_{ij}^2.$$

Then exactly

$$\boxed{q_{ij}^+ = q_{ij} - 2\eta a_{ij} s_{ij} + \eta^2 s_{ij}^2.}$$

Random matrix theory closes the dynamics

Define deterministic kernels by

$$\mathbb{E}[a_{ij}s_{ij} \mid \Delta] \simeq \frac{\mathfrak{d}_{ij}}{\sqrt{\mathcal{R}}} q_{ij}, \qquad \mathbb{E}[s_{ij}^2 \mid \Delta] \simeq \mathfrak{v}_{ij}.$$

Then

$$\boxed{q_{ij}(t+1) = q_{ij}(t) - \frac{2\eta\mathfrak{d}_{ij}(t)}{\sqrt{\mathcal{R}(t)}} q_{ij}(t) + \eta^2 \mathfrak{v}_{ij}(t),}$$

with

$$\mathcal{R}(t) = \frac{1}{2} \sum_{i,j} \mu_i \lambda_j q_{ij}(t).$$

The random-matrix problem is to compute $(\mathfrak{d}_{ij}, \mathfrak{v}_{ij})$ from the two covariances and B.

The gradient Gram matrix is double Wishart

First decouple residuals from the Gaussian data:

$$G_0 = \frac{1}{B} Y D X^\top, \qquad H_0 = G_0 G_0^\top.$$

Rewrite

$$H_0 = \frac{1}{B} Y C Y^\top, \qquad C = \frac{1}{B} D X^\top X D.$$

Conditioned on X, the outer matrix is an ordinary sample covariance matrix with population matrix C.

One Marchenko–Pastur computation is nested inside another.

When $D = I$ and the dimensions are square, the resulting spectrum is the order-two Fuss–Catalan law.

Matrix Stein's lemma avoids entrywise bookkeeping

Condition on C and set

$$\widehat{H}_0 = \Lambda^{1/2} W W^\top \Lambda^{1/2}, \qquad \widehat{R} = (\widehat{H}_0 - zI)^{-1}.$$

For an independent Gaussian direction $\widetilde{W}$,

$$\boxed{\mathbb{E}[F(W) W^\top \mid C] = \mathbb{E}[(\partial_{\widetilde{W}} F(W)) \widetilde{W}^\top \mid C].}$$

The two directional derivatives are

$$\partial_{\widetilde{W}} \widehat{H}_0 = \Lambda^{1/2}(\widetilde{W} W^\top + W \widetilde{W}^\top)\Lambda^{1/2}, \qquad \partial_{\widetilde{W}} \widehat{R} = -\widehat{R}(\partial_{\widetilde{W}} \widehat{H}_0)\widehat{R}.$$

Coherent contractions create the deterministic trace

Apply matrix Stein to $F(W) = \widehat{R}\Lambda^{1/2}W$.

The independent direction produces two contractions:

$$\mathbb{E}_{\widetilde{W}}[\widetilde{W}A\widetilde{W}^\top] = \frac{1}{p}\operatorname{tr}(A)I \quad \text{(coherent)},$$

while the crossed contraction is an explicit p^{-1} matrix correction. The master equation is

$$\boxed{\mathbb{E}[\widehat{R}\widehat{H}_0 \mid C] = \mathbb{E}\left[q_p(z)\widehat{R}\Lambda - \frac{1}{p}\widehat{R}\widehat{H}_0\widehat{R}\Lambda \mid C\right],}$$

where

$$q_p(z) = 1 - \frac{1}{p}\operatorname{tr}(\widehat{R}\widehat{H}_0).$$

The outer fixed point is a deformed MP equation

The incoherent correction vanishes at leading order, giving

$$\widehat{R}(z) \simeq (\gamma_{\mathrm{out}} q(z) C - zI)^{-1}.$$

Therefore

$$\boxed{q(z) = \left[1 + \int \frac{t}{\gamma_{\mathrm{out}} q(z) t - z}\, d\nu_C(t)\right]^{-1}.}$$

The normalized resolvent trace is

$$m_{H_0}(z) = \frac{1}{\gamma_{\mathrm{out}}} \int \frac{d\nu_C(t)}{\gamma_{\mathrm{out}} q(z) t - z} + \frac{\gamma_{\mathrm{out}}^{-1} - 1}{z}.$$

The inner law ν_C comes from the same calculation applied to $C = DX^\top XD/B$.

Anisotropy introduces two weighted traces

Set

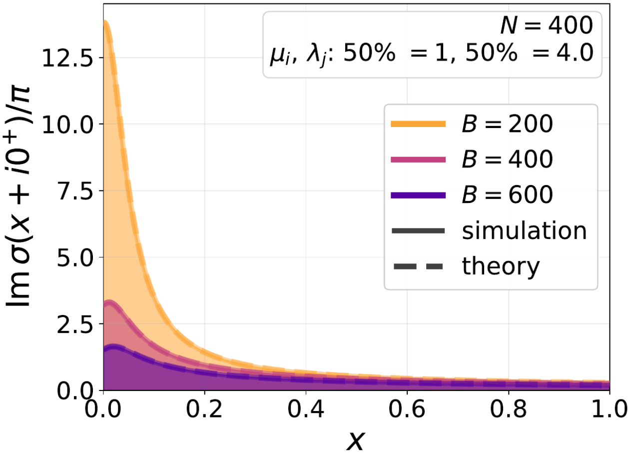
$$\sigma = \frac{1}{B}\,\mathrm{tr}(R\Sigma_{\mathrm{out}}), \qquad \widetilde{\sigma} = \frac{1}{B}\,\mathrm{tr}(\widetilde{R}\Sigma_{\mathrm{in}}).$$

The fixed point is

$$\boxed{\begin{aligned} \sigma &= \frac{1}{B}\sum_i \frac{\mu_i}{\widetilde{s}_1\mu_i - z}, & \widetilde{\sigma} &= \frac{1}{B}\sum_j \frac{\lambda_j}{s_1\lambda_j - z}, \\ s_1 &= \widetilde{s}_1\frac{\sigma}{\widetilde{\sigma}}, & \widetilde{s}_1\sigma &= f(\xi), \\ \xi &= (-2z\sigma\widetilde{\sigma})^{-1/2}. \end{aligned}}$$

Here $f(\xi) = 1 - \sqrt{\pi}\,\xi e^{\xi^2}\,\mathrm{erfc}(\xi)$ records the limiting residual-weight distribution.

The fixed point predicts the batch spectrum



The fixed-point density matches Monte Carlo on both sides of the batch threshold.

The spectrum is not the descent

The normalized gradient has a residual diagonal

$$\frac{G}{\sqrt{2\mathcal{R}}} = \frac{1}{B} Y \widehat{D} X^{\top}.$$

Replacing $\widehat{D}$ by an independent diagonal with the same empirical law does not change the leading bulk spectrum.

But descent requires the overlap with the current error:

$$\boxed{\mathbb{E}[R(z)G \mid \Delta] \simeq c(z)\, R_{\mathrm{eq}}(z)\Sigma_{\mathrm{out}}\Delta\Sigma_{\mathrm{in}}\widetilde{R}_{\mathrm{eq}}(z).}$$

The dependence between residuals and data is precisely what makes this observable nonzero.

The same resolvent gives the descent coefficient

For a spectral transform φ,

$$\varphi(H)G = -\frac{1}{2\pi i}\oint \varphi(z)R(z)G\,dz.$$

Thus

$$\boxed{\mathfrak{d}_{ij} = -\frac{2\sqrt{\mathcal{R}}\,\mu_i\lambda_j}{\pi}\int_0^\infty x\,\varphi(2\mathcal{R}x)\,\mathrm{Im}\left[\frac{\xi^2 f(\xi)}{(\widetilde{s}_1\mu_i - z)(s_1\lambda_j - z)}\right]_{z=x+i0^+} dx.}$$

For SignSVD, $\varphi(s) = s^{-1/2}$, so the current-risk dependence cancels.

Isotropy collapses every paired mode

Take

$$\Sigma_{\mathrm{in}} = \Sigma_{\mathrm{out}} = I_N.$$

All N^2 modes are exchangeable:

$$\mathcal{R}(t+1) = \mathcal{R}(t) - 2\eta\mathfrak{d}\sqrt{\mathcal{R}(t)} + \frac{\eta^2}{2}\mathfrak{v}.$$

algorithm	drift $\mathfrak{d}$	volatility $\mathfrak{v}$
SignSVD	$C(N/B)$	$\min\{B, N\}$
SignSGD	$\mathcal{N}_B/\sqrt{\pi}$	N^2

Here $\mathcal{N}_B \sim \sqrt{B}$ and $C(N/B)$ is a spectral moment of the fixed-point density.

SignSGD drift is controlled by density at zero

Separate one entry's small signal from the diffuse residual:

$$G_{pn} \approx \frac{1}{B} \sum_{b=1}^{B} \left[y_{bp}^2 x_{bn}^2 \Delta_{pn} + y_{bp} x_{bn} \sqrt{2\mathcal{R}}\, z_b \right].$$

A local limit theorem gives density convergence of the second term to $Z \sim \mathcal{N}(0, s^2)$. For small μ, $\mathbb{E}[\mathrm{sign}(Z + \mu)] = \sqrt{2/\pi}\, \mu/s + O((\mu/s)^3)$, so

$$\boxed{\mathfrak{d}_{pn}^{\mathrm{SignSGD}} = \frac{\mathcal{N}_B}{\sqrt{\pi}} \sim \sqrt{\frac{B}{\pi}}.}$$

SignSGD volatility is coordinatewise

Each entry of $S = \mathrm{sign}(G)$ is ± 1, so

$$\boxed{s_{pn}^2 = 1, \qquad \|S\|_F^2 = N^2.}$$

$$\eta_\star(t) = \frac{2\mathfrak{d}\sqrt{\mathcal{R}(t)}}{\mathfrak{v}}, \qquad \frac{\eta_\star^{\mathrm{SignSVD}}}{\eta_\star^{\mathrm{SignSGD}}} \asymp \begin{cases} \sqrt{N}, & B \geq N, \\ N/\sqrt{B}, & B \leq N. \end{cases}$$

Their learning-rate scales differ even when tuned contractions agree.

Isotropy removes the reason to precondition

After tuning each method to its natural learning-rate scale,

$$\text{SignSVD contraction} \asymp \text{SignSGD contraction}$$

up to absolute constants.

Muon's spectral geometry becomes useful only when the covariance is anisotropic.

This sanity check prevents a normalization difference from being mistaken for an algorithmic advantage.

Half-anisotropy turns N^2 paired modes into N row norms

One isotropic side removes one spectral index

Set

$$\Sigma_{\mathrm{in}} = I_N, \qquad \Sigma_{\mathrm{out}} = U \, \mathrm{diag}(\mu_1, \ldots, \mu_N) U^\top.$$

Define the row-projected risks

$$\boxed{r_i(t) = \sum_{j=1}^{N} q_{ij}(t) = \|\Delta_t^\top u_i\|^2, \qquad \mathcal{R}(t) = \frac{1}{2} \sum_i \mu_i r_i(t).}$$

They satisfy

$$r_i(t+1) = r_i(t) - \frac{2\eta \mathfrak{d}_i}{\sqrt{\mathcal{R}(t)}} r_i(t) + \eta^2 \mathfrak{v}_i.$$

SignSVD volatility is a projection

Proposition (SignSVD volatility is a projection)

Let $S = \mathrm{polar}(G) = U_r V_r^\top$. *Then*

$$\boxed{\|S^\top u_i\|^2 = u_i^\top P_{\mathrm{col}(G)} u_i.}$$

If $B \geq N$ *and* G *has full row rank, this equals one for every mode. If* $B < N$ *and* $\mathrm{rank}(G) = B$, *then* $0 \leq \mathfrak{v}_i \leq 1$ *and* $\sum_i \mathfrak{v}_i = B$.

Orthogonalization cancels the singular values

Because $S = U_r V_r^\top$,

$$SS^\top = U_r V_r^\top V_r U_r^\top = U_r U_r^\top = P_{\mathrm{col}(G)}.$$

Therefore

$$\|S^\top u_i\|^2 = u_i^\top SS^\top u_i.$$

For a general spectral map,

$$SS^\top = U_r \operatorname{diag}\left(\varphi(\sigma_k^2)^2 \sigma_k^2\right) U_r^\top,$$

so the volatility keeps a nontrivial spectral weight.

The map $\sigma_k \mapsto 1$ turns a difficult variance calculation into a projector identity.

Half-anisotropy leaves one population spectral sum

Set $\gamma = N/B$. The reduced fixed point is

$$\sigma = \frac{1}{B}\sum_i \frac{\mu_i}{\widetilde{s}_1\mu_i - z}, \quad \widetilde{s}_1\sigma = f(\xi),$$

$$\widetilde{\sigma} = \frac{f(\xi)-\gamma}{z}, \qquad s_1 = z + \frac{\gamma}{\widetilde{\sigma}}.$$

$$\boxed{\begin{aligned} \rho_i(x) &= \frac{1}{\pi}\,\mathrm{Im}\,\frac{1}{\widetilde{s}_1(x+i0^+)\mu_i - x}, \\ \mathfrak{d}_i &= \frac{1}{\gamma}\int_0^\infty \sqrt{x}\,\rho_i(x)\,dx, \\ \mathfrak{v}_i &= \int_0^\infty \rho_i(x)\,dx. \end{aligned}}$$

Large batches produce square-root drift

For $B \geq N$,

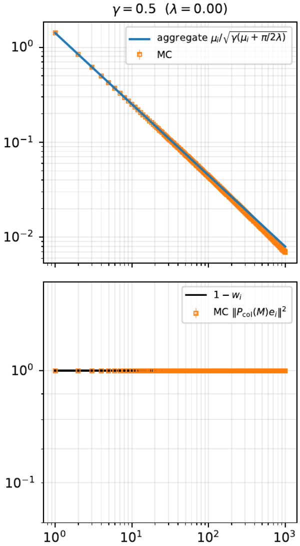
$$\boxed{\mathfrak{d}_i = \sqrt{\frac{\mu_i}{\gamma}}, \qquad \mathfrak{v}_i = 1.}$$

In algorithmic time,

$$\frac{dr_i}{d\tau} = -2\sqrt{\frac{\mu_i}{\gamma}}\, r_i.$$

Thus

$$\text{GD:} \; -\mathcal{K}(\Delta), \qquad \text{SignSVD:} \; -\mathcal{K}^{1/2}(\Delta).$$



Small batches create a spectral resolution boundary

Define Λ by

$$B = \sum_i \frac{\Lambda\mu_i}{1+\Lambda\mu_i}.$$

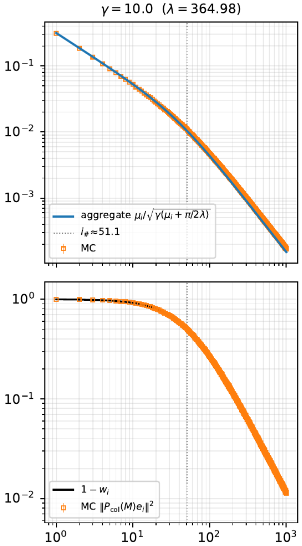
Then

$$\mathfrak{v}_i = \frac{\Lambda\mu_i}{1+\Lambda\mu_i},$$

and

$$\mathfrak{d}_i \approx \begin{cases} \sqrt{\mu_i}\sqrt{B/N}, & \Lambda\mu_i \gg 1, \\ \mu_i\sqrt{2\Lambda B/(\pi N)}, & \Lambda\mu_i \ll 1. \end{cases}$$

The head is preconditioned; the unresolved tail returns to SGD-like drift.



Power laws set the number of resolved modes

Take

$$\mu_i = i^{-\alpha}, \qquad r_i(0) = i^{-\beta}, \qquad \alpha + \beta > 1.$$

The boundary $i_\# = \Lambda^{1/\alpha}$ obeys

$$\boxed{i_\# \asymp \begin{cases} B, & \alpha > 1, \\ B/\log(N/B), & \alpha = 1, \\ B^{1/\alpha}/N^{1/\alpha - 1}, & 0 < \alpha < 1. \end{cases}}$$

For $\alpha = \frac{1}{2}$, $i_\# \asymp B^2/N$: a batch of order $\sqrt{N}$ resolves only order-one modes.

The batch selects between two spectral regimes

mode range	SignSVD drift	SignSVD volatility
resolved head	$\sqrt{\mu_i}\sqrt{B/N}$	1
unresolved tail	$\mu_i\sqrt{2\Lambda B/(\pi N)}$	$\Lambda\mu_i$

For comparison, half-anisotropic SignSGD retains

$$\mathfrak{d}_i^{\mathrm{SignSGD}} = \frac{\mu_i \mathcal{N}_B}{\sqrt{\pi\overline{\mu}}}, \qquad \overline{\mu} = \frac{1}{N}\sum_i \mu_i.$$

Batch size is a spectral resolution scale, not only a variance-reduction parameter.

Compare algorithms at a common noise floor

For kernels $(\mathfrak{d}_i, \mathfrak{v}_i)$, define

$$\mathcal{S} = \sum_i \frac{\mu_i \mathfrak{v}_i}{2\mathfrak{d}_i}, \qquad \boxed{\eta^\star = \frac{2\sqrt{\epsilon}}{\mathcal{S}}.}$$

Use this single constant learning rate throughout the run and measure

$$t_{4\epsilon} = \inf\{t \geq 0 : \mathcal{R}(t) \leq 4\epsilon\}.$$

This avoids comparing two optimizers at the same numerical learning rate even though their update norms differ by powers of N.

Square-root drift changes the time-to-accuracy exponent

Theorem (P.–Marshall–Benigni–Wang–Agarwala–P., 2026)

Let $\gamma = N/B \leq 1$, $\mu_i = i^{-\alpha}$, $r_i(0) = i^{-\beta}$, and $\alpha + \beta > 1$. Then

$$t_{4\epsilon}^{\mathrm{SignSVD}} \asymp \begin{cases} \gamma N^{1-\alpha/2} \epsilon^{-\alpha/[2(\alpha+\beta-1)]}, & \beta < 1, \\ \mathcal{S}_{\mathrm{SignSVD}} \epsilon^{-1/2}, & \beta > 1, \end{cases}$$

and

$$t_{4\epsilon}^{\mathrm{SignSGD}} \asymp \begin{cases} \dfrac{\pi N^2 \overline{\mu}}{B} \epsilon^{-\alpha/(\alpha+\beta-1)}, & \beta < \alpha + 1, \\ \mathcal{S}_{\mathrm{SignSGD}} \epsilon^{-1/2}, & \beta > \alpha + 1. \end{cases}$$

The first SignSVD branch assumes $0 < \alpha < 2$; boundaries have logarithmic corrections.

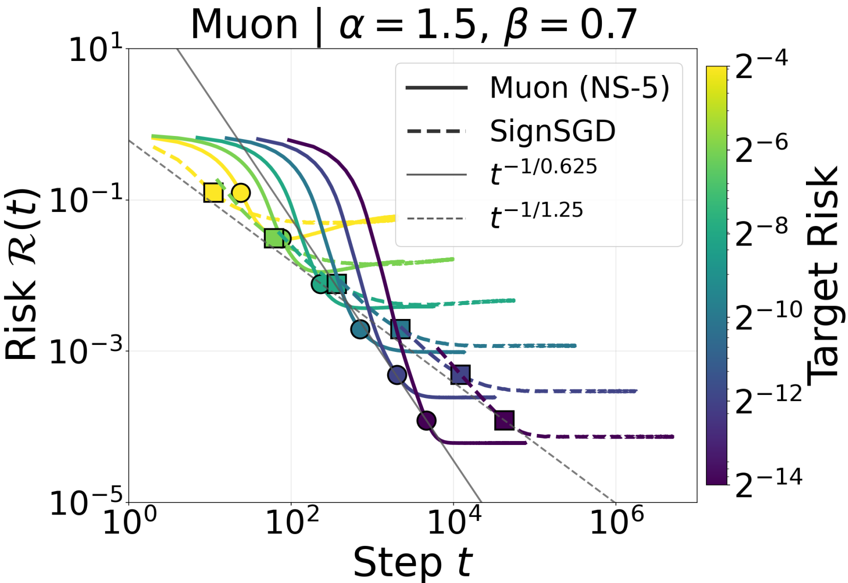
Target structure decides whether orthogonalization helps

target regime	range	comparison
tail-heavy	$\beta < 1$	SignSVD is polynomially faster as $\epsilon \to 0$
intermediate	$1 < \beta < \alpha + 1$	SignSGD may lead early; SignSVD wins at high accuracy
head-heavy	$\beta > \alpha + 1$	both use the $\epsilon^{-1/2}$ branch

The mechanism is spectral:

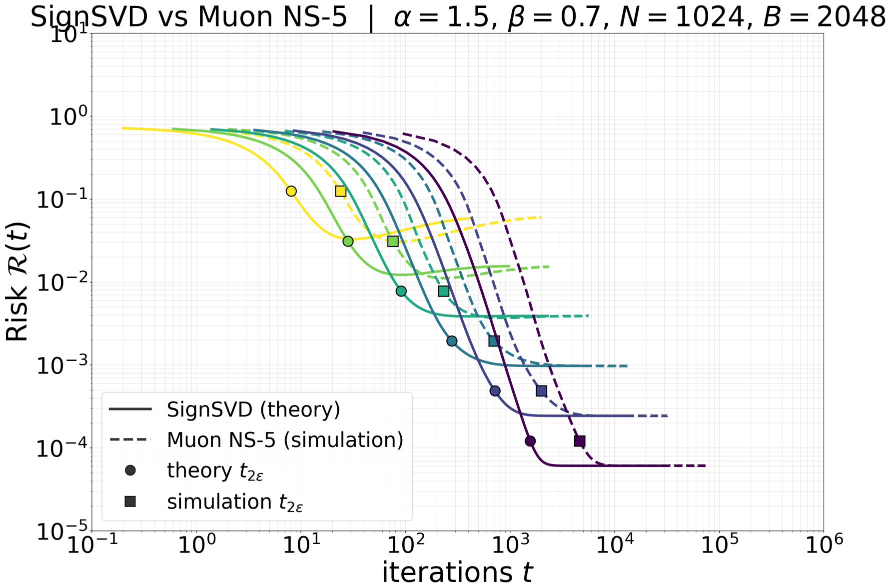
$$\mathfrak{d}_i^{\mathrm{SignSVD}} \propto \mu_i^{1/2}, \qquad \mathfrak{d}_i^{\mathrm{SignSGD}} \propto \mu_i.$$

Tail-heavy targets expose the different exponents



Momentum is a separate temporal question

Five Newton–Schulz steps already track SignSVD



Momentum inflates integrated volatility

Suppose $\mathrm{Corr}(\xi_t, \xi_{t+k}) \approx \beta_{\mathrm{mom}}^k$. Then

$$\boxed{\begin{aligned} \mathrm{Var}\left(\sum_{t=1}^{T} \xi_t\right) &\approx Tv\left(1 + 2\sum_{k\geq 1} \beta_{\mathrm{mom}}^k\right) \\ &= Tv\,\frac{1+\beta_{\mathrm{mom}}}{1-\beta_{\mathrm{mom}}}, \\ \text{SD inflation} &= \sqrt{\frac{1+\beta_{\mathrm{mom}}}{1-\beta_{\mathrm{mom}}}}. \end{aligned}}$$

At fixed learning rate, momentum raises the noise floor

Define $\mathcal{N}_{\text{eff}}$ by $\mathcal{R}_\infty \approx (\eta \mathcal{N}_{\text{eff}}/2)^2$.

β_{mom}	measured $\mathcal{N}_{\text{eff}}$	AR(1) prediction
0	9.97	baseline
0.90	36.90	36.59
0.95	52.49	52.43
0.99	117.33	118.42

Fixed learning rate

Each run fixes η; reaching floor ϵ requires $\eta^\star = 2\sqrt{\epsilon}/\mathcal{N}_{\text{eff}}$ (model-specific).

Selected references

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Three mechanisms explain the spectral gain

Matrix structure

$$G_B = \frac{1}{B}\sum_b d_b(y_b \otimes x_b)$$

The batch gradient is low rank and has two covariance geometries.

Preconditioning

$$\mu_i \longmapsto \sqrt{\mu_i}$$

Resolved modes receive square-root covariance drift.

Resolution

$$\sum_i \mathfrak{v}_i = B$$

The batch determines which modes are preconditioned.

Muon's matrix operation matters when anisotropy and target mass make weak covariance directions the optimization bottleneck.